Dashboard Design

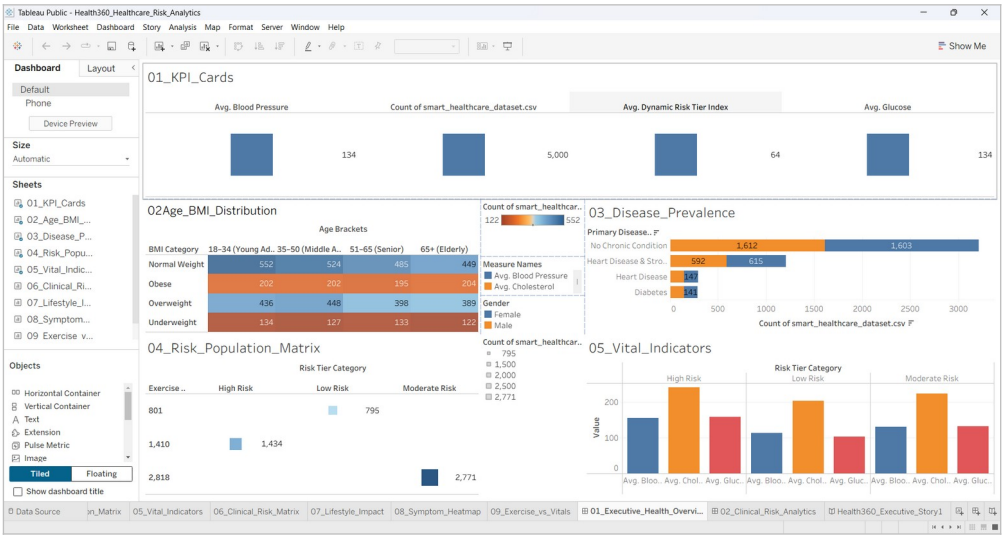
Date	28 October 2026
Team ID	[Insert Team ID]
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	5 Marks

Activity 1: Dashboard Design System & Styling Rules

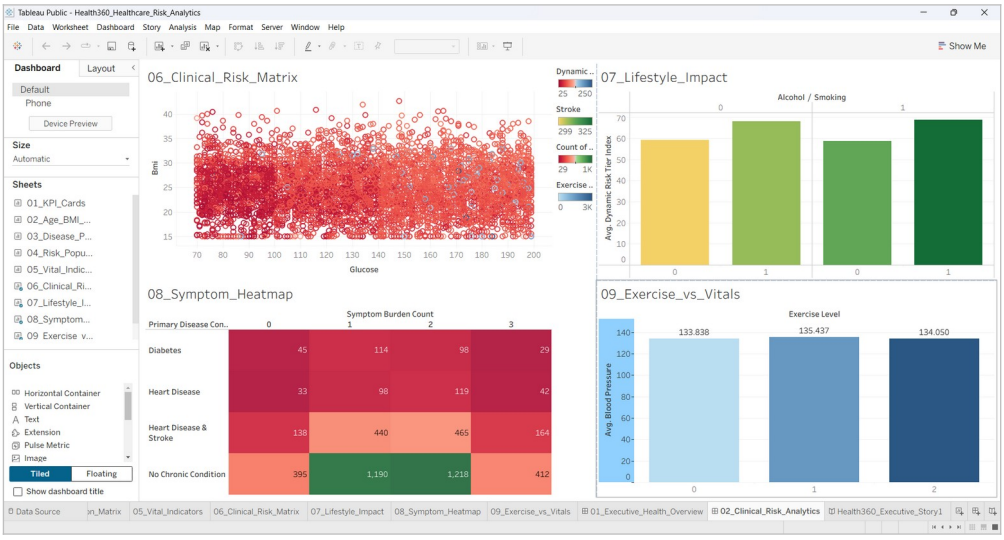
The Health360 dashboards follow a single design system so that hospital executives and clinical risk managers can read population-level KPIs and patient-level risk signals at a glance. The same card structure, colour logic and typography are used across the Executive Health Overview, the Clinical Risk Analytics dashboard and the embedded executive story.

- **Layout & Structure:** Grid-based responsive card design that separates high-level executive KPIs from granular scatter plots and symptom heatmaps.
- **Executive Primary Colours:** Navy Blue (**#1E293B**) for headings and primary elements on a clean card background (**#F8FAFC**).
- **High Risk Tier:** Crimson Red (**#DC2626**) for a composite risk index of 75 or above, driven mainly by blood pressure above 140 mmHg or glucose above 140 mg/dL.
- **Moderate Risk Tier:** Amber Gold (**#D97706**) for an index of 50–74, with blood pressure above 120 mmHg and glucose above 100 mg/dL scoring in the elevated band.
- **Low Risk Tier:** Emerald Green (**#16A34A**) for an index below 50.
- **Typography:** Arial / Segoe UI, a clean sans-serif chosen for legibility across devices.

Activity 2: Executive Dashboard 1 & Clinical Risk Dashboard 2 Outcomes



Dashboard 1: 01_Executive_Health_Overview



Dashboard 2: 02_Clinical_Risk_Analytics

Here are the five primary outcomes from the Health360 dashboard system:

- **Overweight and obese categories account for nearly half of all patients across every age group**: The BMI distribution heatmap places 2,474 of the 5,000 patient records (49.5%) in the Overweight or Obese categories, and the share holds at roughly 48–51% in all four age brackets.
- **High-risk patient clustering correlates strongly with elevated blood pressure and glucose levels**: The Risk Population Matrix shows 1,434 High Risk patients (28.7%), and among patients with both blood pressure and glucose above 140, about 86% fall in the High Risk tier, making these two vitals the key threshold drivers.
- **Co-occurring Heart Disease and Stroke is the largest chronic condition group**: The Symptom Burden Heatmap shows 1,207 patients (about 24%) with both conditions, most of them carrying one or two symptoms (440 and 465 patients), which helps clinical managers triage resources for multi-condition patients.

- **Smoking is the strongest lifestyle driver of high-risk classification**: The Lifestyle Impact chart shows an average risk index of about 59 for non-smokers against about 68 for smokers, and 39% of smokers fall in the High Risk tier versus 18% of non-smokers (roughly 2.2x), while alcohol use alone adds little.
- **Guided executive story streamlines strategic clinical decision-making**: The embedded four-scene Health360 Executive Story moves leadership from population overview and condition prevalence, through risk tiers and clinical risk factors, to disease symptom burden analysis, supporting prioritisation of interventions and resources.

Overall Outcome: The Health360 dashboard ecosystem shows that chronic disease risk is driven by a combination of elevated vital thresholds (blood pressure and glucose), weight status, smoking and multi-condition symptom burdens, providing an actionable framework for preventive healthcare.

Activity 3: Web Embedding & Container Integration

- **Host Environment:** Lightweight Python Flask web application (app.py) serving responsive HTML5/CSS3 pages locally on http://127.0.0.1:5001.
- **Integration Layer:** Tableau JavaScript API v2 enabling interactive filtering, dynamic tab switching and story scene transitions directly from web controls.

Figure 1: Health360 Master Dashboard Architecture & Web Execution Interface

